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The Physics Behind the Warehouse Optimizer

From Statistical Mechanics to Combinatorial Optimization

Abstract

This document provides a detailed exposition of the physical principles underpinning the warehouse layout optimizer implemented in C. The algorithm draws directly from statistical mechanics and thermodynamics: it formulates the warehouse configuration as a classical many-body system with a Hamiltonian energy function, introduces repulsive and attractive inter-bay potentials analogous to molecular forces, and searches for the global energy minimum using Simulated Annealing — a stochastic process that mimics the thermodynamic cooling of a physical system. We explain each physical concept — the Hamiltonian, micro- and macrostates, the Boltzmann distribution, temperature schedules, and gravitational attraction — as they appear concretely in the source code.

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1 Introduction: The Physical Analogy

Placing storage bays in a warehouse is a combinatorial optimization problem: the number of possible configurations grows super-exponentially with the number of bays. Exhaustive search is computationally impossible. Instead, the optimizer exploits a powerful analogy with statistical physics.

In condensed-matter physics, a system of N interacting particles seeks its lowest-energy configuration (the ground state) by exchanging energy with a thermal reservoir. As the temperature T is lowered slowly, the system anneals — it avoids local traps and settles into the global minimum. This is precisely the strategy exported to the optimization domain by **Simulated Annealing** (Kirkpatrick et al., 1983).

The physical correspondence is:

Physics	Optimizer
Particle configuration	Bay layout $\{b_i\}$
Hamiltonian $\mathcal{H}$	Energy function $E(\{b_i\})$
Ground state	Optimal layout
Thermal fluctuation	Random perturbation (birth / death / move)
Temperature T	Acceptance parameter <code>temp</code>
Annealing schedule	$T \leftarrow \alpha T$, $\alpha = 0.9998$
Boltzmann factor	$e^{-\Delta E/T}$
Microstate	A specific $(x_i, y_i, \theta_i, \text{id}_i)$ tuple
Macrostate	Energy level and bay count

2 The Hamiltonian: Defining the Energy of the System

2.1 General Form

In classical mechanics, the Hamiltonian $\mathcal{H}$ gives the total energy of a system. Here, the “system” is the set of N bays placed in the warehouse. There are no kinetic terms (bays do not move in real time), so the Hamiltonian is purely a potential energy:

$$\mathcal{H}(\{b_i\}) = \mathcal{H}_{\text{score}} + \mathcal{H}_{\text{area}} + \mathcal{H}_{\text{attr}} + \mathcal{H}_{\text{penalty}} \quad (1)$$

Each term is computed inside the function `calcular_energia_global()`.

2.2 Score Term: Efficiency Reward

The optimizer wants to maximize the storage efficiency of placed bays. This is encoded as a *negative* energy contribution (the system gains energy by being efficient, and the optimizer minimizes energy):

$$\mathcal{H}_{\text{score}} = - \sum_{i=1}^N \eta_i \quad (2)$$

where $\eta_i = P_i/L_i$ is the efficiency of bay i (total weight capacity P_i divided by the number of levels L_i). In the code:

```
score_total += bt->eff; // bt->eff = P / nL
```

and the function returns $E = -\text{score_total} + \dots$ (subtracting from energy means higher efficiency lowers energy).

2.3 Area Penalty Term: Coverage Cost

A secondary term penalizes wasteful area usage linearly:

$$\mathcal{H}_{\text{area}} = +\lambda \sum_{i=1}^N A_i, \quad \lambda = 0.005 \quad (3)$$

where $A_i = W_i \cdot D_i$ is the footprint of bay i . This acts as a “pressure” preventing the system from filling the warehouse indiscriminately with low-efficiency bays.

3 Inter-Bay Potentials: Attraction and Repulsion

3.1 Gravitational Attraction Potential

The most physically rich term in the Hamiltonian is a pairwise attractive potential between bays, modeled after Newtonian gravity modulated by a Gaussian (reminiscent of the Lennard-Jones potential in its attractive tail):

$$\mathcal{H}_{\text{attr}} = -G \sum_{i < j} \frac{A_i A_j}{10^8} \exp\left(-\frac{r_{ij}^2}{2\sigma^2}\right) \quad (4)$$

where:

- $G = 17.0$ is the “gravitational constant” of the system (tuned empirically).
- A_i, A_j are the footprint areas of bays i and j (acting as masses).
- $r_{ij}^2 = (x_i - x_j)^2 + (y_i - y_j)^2$ is the squared Euclidean distance between bay centres.
- $\sigma = 25\,000\,000$ is the characteristic length scale of the attraction (in distance units squared inside the exponent).

In the code:

```
double attraction = (areas[i]*areas[j] / 1e8)
                  * exp(-dist_sq / (2.0*SIGMA*SIGMA));
u_attr -= K_GRAVITY * attraction;
```

Physical interpretation. Larger bays attract each other more strongly (mass proportional to area). Nearby bays attract most; the Gaussian envelope damps the interaction at large separations. This *clustering force* pushes bays to pack together compactly, which is physically desirable — a dense warehouse maximizes utilization.

3.2 Hard-Wall Repulsion: Collision Penalties

While attraction organises the layout, overlapping physical objects is forbidden. Collisions are penalised with a *hard-wall repulsive potential*:

$$\mathcal{H}_{\text{penalty}}^{\text{collision}} = \Phi_{\infty} \sum_{i < j} \mathbb{I}[\text{bay}_i \cap \text{bay}_j \neq \emptyset] \quad (5)$$

where $\Phi_{\infty} = 10^{15}$ (coded as `BIG_PEN = 1016`). This is the discrete analogue of the infinitely steep repulsive core in a Lennard-Jones or hard-sphere potential. Any

configuration with overlapping bays has an astronomically high energy and is rejected with virtual certainty.

In the code:

```
if (paths_intersect_fast(&racks[i], &racks[j]) || ...) {
    total_pen += BIG_PEN;
    pen_indiv[i] += 1e15;
}
```

3.3 Boundary and Obstacle Penalties

Bays that violate the warehouse boundary or intersect obstacles receive a similar hard-wall penalty:

$$\mathcal{H}_{\text{penalty}}^{\text{boundary}} = \Phi_{\infty} \mathbb{I}[\text{bay not contained in warehouse}] \quad (6)$$

$$\mathcal{H}_{\text{penalty}}^{\text{obstacle}} = \Phi_{\infty} \mathbb{I}[\text{bay intersects any obstacle}] \quad (7)$$

These constraints implement the physical equivalent of impenetrable walls: the warehouse polygon and fixed obstacles are simply regions of infinite potential energy that the system cannot enter.

3.4 Height (Ceiling) Constraint

A ceiling profile $h(x)$ is given as a piecewise-constant function. A bay of height H_i centred at (x_i, y_i) must satisfy:

$$H_i \leq h_{\min}(x_i^-, x_i^+) \quad (8)$$

where $h_{\min}$ is the minimum ceiling height over the bay's horizontal extent. Violation again costs Φ_{∞} .

4 Microstates and Macrostates

4.1 Defining Microstates

A **microstate** of the system is a complete, fully-specified configuration of all bays:

$$\mu = \{(x_i, y_i, \theta_i, \text{id}_i)\}_{i=1}^N \quad (9)$$

Each microstate has a well-defined energy $E(\mu) = \mathcal{H}(\mu)$. The optimizer explores the space of microstates through random perturbations.

4.2 Defining Macrostates

A **macrostate** is characterised by macroscopic observables — quantities that many microstates share. Here the most natural macrostates are labelled by:

- The total number of bays N .
- The total energy E .
- The total covered area A_{cov} .

Many distinct microstates (different positions, rotations, or bay assignments) can correspond to the same macrostate label. The **degeneracy** $\Omega(E, N)$ counts how many microstates share energy E with N bays. In statistical mechanics, entropy is $S = k_B \ln \Omega$; here the optimizer implicitly accounts for this degeneracy through the Boltzmann acceptance criterion.

4.3 Phase Space

The full phase space Γ is the set of all valid microstates:

$$\Gamma = \bigcup_{N=0}^{N_{\max}} (\mathcal{X} \times \mathcal{Y} \times \Theta \times \text{ID})^N \quad (10)$$

where $\mathcal{X}, \mathcal{Y}$ are continuous coordinate ranges, $\Theta = \{0, \pi/2, \pi, 3\pi/2\}$ in normal mode (discrete), and ID is the set of bay types. The optimizer performs a random walk on Γ .

5 Simulated Annealing: The Boltzmann Distribution

5.1 The Canonical Ensemble

In statistical mechanics, a system in contact with a heat bath at temperature T occupies microstate μ with probability given by the **Boltzmann distribution**:

$$P(\mu) = \frac{e^{-\mathcal{H}(\mu)/k_B T}}{Z}, \quad Z = \sum_{\mu'} e^{-\mathcal{H}(\mu')/k_B T} \quad (11)$$

where Z is the **partition function**. At high T , all microstates are nearly equally probable (the system explores broadly). As $T \rightarrow 0$, the distribution concentrates entirely on the ground state (minimum energy configuration).

Simulated Annealing mimics this process computationally. Setting $k_B = 1$ (absorbed into T), the algorithm:

1. Proposes a new microstate μ' from the current μ .
2. Accepts μ' with probability:

$$P_{\text{accept}}(\mu \rightarrow \mu') = \begin{cases} 1 & \text{if } \Delta E \leq 0 \\ e^{-\Delta E/T} & \text{if } \Delta E > 0 \end{cases} \quad (12)$$

This is the **Metropolis-Hastings criterion**, and it guarantees that the stationary distribution of the Markov chain is the Boltzmann distribution (11) at temperature T .

In the code:

```
if (e_nueva < e_actual) {
    acepta = 1;
} else {
    double prob = exp(-fabs(e_nueva - e_actual) / temp);
    acepta = (rand_uniform(0, 1) < prob);
}
```

5.2 Free Energy and the Optimizer’s True Objective

At finite temperature, the system minimizes not the internal energy E but the **Helmholtz free energy**:

$$F = E - TS = E - T k_B \ln \Omega(E) \quad (13)$$

The entropy term $-TS$ biases the optimizer toward macrostates with high degeneracy (many equivalent configurations), which helps escape narrow local minima. This is precisely why accepting uphill moves is beneficial: it grants access to entropic regions of phase space that eventually lead to better ground states.

6 The Temperature Schedule: Annealing

6.1 Cooling Law

The code uses **geometric cooling** (also called exponential annealing):

$$T_{k+1} = \alpha T_k, \quad \alpha = 0.9998, \quad T_0 = 10\,000 \quad (14)$$

After n iterations:

$$T_n = T_0 \cdot \alpha^n = 10^4 \cdot (0.9998)^n \quad (15)$$

At $n = 100\,000$ iterations (the total budget), the final temperature is:

$$T_{100000} = 10^4 \cdot (0.9998)^{100000} \approx 10^4 \cdot e^{-20} \approx 2.06 \times 10^{-5} \quad (16)$$

This enormous dynamic range — from $T = 10^4$ (highly exploratory) to $T \approx 10^{-5}$ (essentially deterministic) — replicates the annealing of a physical material from a molten state to a crystalline ground state.

6.2 Physical Interpretation of Temperature Stages

Stage	Temperature range	Physical analogy
Early	$T \sim 10^3\text{--}10^4$	Molten liquid — high mobility, global exploration
Mid	$T \sim 10^1\text{--}10^2$	Supercooled liquid — local rearrangements
Late	$T \sim 10^{-3}\text{--}10^0$	Near-crystalline — fine-tuning, convergence

At high temperature, the exponential $e^{-\Delta E/T}$ is close to 1 for any ΔE , so uphill moves are freely accepted. As T drops, only smaller uphill moves are accepted, until eventually the algorithm performs steepest descent.

7 Perturbation Dynamics: Transitions Between Microstates

7.1 The Move Set

The Markov chain transitions between microstates via three types of perturbation, selected with probabilities that depend on the current state:

Birth (probability ≈ 0.65). A new bay is added at a random position:

$$\mu \rightarrow \mu \cup \{(x_{\text{rand}}, y_{\text{rand}}, \theta_{\text{rand}}, \text{id}_{\text{rand}})\} \quad (17)$$

This changes the particle number $N \rightarrow N + 1$ — a **grand canonical** move, analogous to a system exchanging particles with a reservoir.

Death (probability ≈ 0.10). The highest-penalty bay is removed:

$$\mu \rightarrow \mu \setminus \{b_{i^*}\}, \quad i^* = \arg \max_i p_i(\mu) \quad (18)$$

where p_i is the individual penalty of bay i . Targeting the worst offender is equivalent to a biased Glauber dynamics in spin systems.

Mutation (probability ≈ 0.25). A bay is displaced or its type is changed:

$$(x_i, y_i) \rightarrow (x_i + \delta x, y_i + \delta y), \quad \delta x, \delta y \sim \mathcal{U}(-60, 60) \quad (19)$$

or the bay type id is resampled uniformly from the available types. This is the discrete analogue of a Langevin step.

7.2 Grand Canonical Ensemble

Since N is variable, the optimizer effectively samples from a **grand canonical ensemble** (GCE). In thermodynamics, the GCE is appropriate for systems that exchange both energy and particles with a reservoir. The grand canonical partition function is:

$$\mathcal{Z} = \sum_{N=0}^{N_{\text{max}}} \sum_{\mu_N} e^{-(\mathcal{H}(\mu_N) - \mu_c N)/T} \quad (20)$$

where μ_c is the chemical potential — the energy cost of adding one bay. In the optimizer this is implicitly encoded: the birth probability (0.65) versus death/mutation probability (0.35) acts as an effective chemical potential that controls the equilibrium bay count.

8 Escape Mechanism: Fine-Angle Mode as a Thermal Spike

8.1 The Problem of Metastable States

At low temperatures, the system can become trapped in a **metastable state** — a local energy minimum with a barrier $\Delta E_{\text{barrier}} \gg T$. The Boltzmann factor $e^{-\Delta E_{\text{barrier}}/T}$ is then negligible, making escape exponentially unlikely.

8.2 Fine-Angle Mode as a Local Temperature Spike

The code implements an escape heuristic: after `FINE_STALL_ITERS` = 4000 consecutive iterations without energy improvement exceeding $\Delta E_{\text{threshold}} = 1.0$, the algorithm enters “fine-angle mode” for `FINE_ACTIVE_ITERS` = 500 iterations.

In fine-angle mode, bay orientations are drawn from 24 discrete angles (multiples of 15) rather than the 4 orthogonal ones. This is physically equivalent to a **local temperature**

spike: the system is briefly heated — its accessible configuration space suddenly expands — allowing it to surmount barriers in the angular degree of freedom.

Mathematically, the angular entropy term changes:

$$S_{\theta}^{\text{ortho}} = \ln 4 \longrightarrow S_{\theta}^{\text{fine}} = \ln 24 \quad (21)$$

The free energy barrier $\Delta F = \Delta E - T\Delta S$ is reduced by $T(\ln 24 - \ln 4) = T \ln 6 \approx 1.79 T$, facilitating escape from angular traps.

After the spike, a **cooldown** of `FINE_COOLDOWN_ITERS` = 2000 iterations prevents immediate re-triggering — analogous to requiring the system to re-equilibrate before a second thermal spike.

9 The Complete Energy Landscape

Combining all terms, the full Hamiltonian (1) can be written explicitly as:

$$\begin{aligned} \mathcal{H}(\{b_i\}) = & - \sum_{i=1}^N \frac{P_i}{L_i} + \lambda \sum_{i=1}^N W_i D_i \\ & - G \sum_{i < j} \frac{W_i D_i \cdot W_j D_j}{10^8} \exp\left(-\frac{r_{ij}^2}{2\sigma^2}\right) \\ & + \Phi_{\infty} \sum_{i < j} \mathbb{K}[\text{overlap}(b_i, b_j)] \\ & + \Phi_{\infty} \sum_i [\mathbb{K}[b_i \notin \mathcal{W}] + \mathbb{K}[b_i \cap \mathcal{O} \neq \emptyset] + \mathbb{K}[H_i > h(b_i)]] \end{aligned} \quad (22)$$

where:

- $\mathcal{W}$ is the warehouse polygon (feasible region).
- $\mathcal{O}$ is the union of obstacle polygons.
- $h(b_i)$ is the effective ceiling height over bay b_i .
- $\lambda = 0.005$, $G = 17.0$, $\sigma = 2.5 \times 10^7$, $\Phi_{\infty} = 10^{16}$.

This landscape has the qualitative structure of a **rugged funnel**: the global minimum is an efficiently packed, non-overlapping, boundary-respecting layout; the hard-wall penalties create infinite barriers around infeasible regions; and the gravitational attraction provides a long-range funnel that guides the system toward compact clusters.

10 Quality Factor Q : The Observable

The final observable computed from the ground-state configuration is the quality factor Q :

$$Q = \left(\frac{\sum_i P_i}{\sum_i L_i} \right)^{2-R} \quad (23)$$

where $R = A_{\text{cov}}/A_{\text{warehouse}}$ is the fractional area coverage ($0 \leq R \leq 1$). When $R \rightarrow 1$ (full coverage), $Q \rightarrow (\bar{P}/\bar{L})^1$; when $R \rightarrow 0$, $Q \rightarrow (\bar{P}/\bar{L})^2$. Thus high coverage reduces the exponent, rewarding efficient use of the warehouse area.

This can be read as a **thermodynamic equation of state**: Q relates the microscopic bay properties (P_i, L_i) to a macroscopic geometric property R , analogous to equations of state like $PV = nRT$ that link microscopic and macroscopic descriptions.

11 Summary

Physics concept	Implementation in code
Hamiltonian $\mathcal{H}$	<code>calcular_energia_global()</code>
Hard-wall repulsion	<code>BIG_PEN = 10¹⁶</code> on overlaps/boundaries
Gravitational attraction	Gaussian pairwise potential in <code>u_attr</code>
Microstate	Bay <code>{id, x, y, theta}</code> tuple
Macrostate	(N, E, A_{cov}) triplet
Grand canonical ensemble	Birth/death moves change N
Boltzmann criterion	<code>exp(-fabs(e_nueva - e_actual) / temp)</code>
Temperature schedule	<code>temp *= 0.9998</code>
Metastability escape	Fine-angle mode (thermal spike in θ -space)
Entropy of angle space	$\ln 4 \rightarrow \ln 24$ in fine mode
Free energy	Implicit in acceptance probability
Ground state	<code>best_layout</code> — minimum energy found
Equation of state	Quality factor $Q = (\bar{P}/\bar{L})^{2-R}$

The warehouse optimizer is, in essence, a classical statistical mechanics simulation of a two-dimensional many-body system with mixed attractive and repulsive interactions, solved via a grand canonical Monte Carlo algorithm (Simulated Annealing) on a slowly cooled thermal schedule.

References

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